Lab 4 — Project: Matrix Completion on a Panel

Machine Learning II · Tutorial

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Goal of the session. The coding project of Homework 3, carried to the end: complete a panel with a staggered treatment by low-rank factorization, read the completed entries as counterfactuals, quantify the uncertainty, and write it up. Self-attention (Problem 18) is the extension for those who finish early. This is the last session of ML II.

Materials

- `homework_3_student.pdf` — Problem 17 (the project) and Problem 18 (self-attention).
- `homework_3_student.ipynb` / `homework_3_solution.ipynb` — Parts 1, 2.
- Lectures 4 and 6 of ML II; ML I, Lecture 6 (Eckart–Young) and Lecture 3 (ridge).

1 The Problem (20 min)

A panel $\mathbf{Y} \in \mathbb{R}^{M\times T}$, $M = 20$ units, $T = 12$ periods. Units adopt a treatment at staggered dates; after adoption, the untreated outcome $Y_{it}(0)$ is missing. The entries we see are the untreated ones, $\Omega = \{(i,t) : \text{not yet treated}\}$. If $\mathbf{Y}(0)$ has low rank — a few common factors drive every unit — then the missing block can be recovered:

$$\min_{\mathbf{U}\in\mathbb{R}^{M\times K},\ \mathbf{V}\in\mathbb{R}^{T\times K}} \frac{1}{2}\sum_{(i,t)\in\Omega}\left(Y_{it} - \mathbf{u}_i^\top \mathbf{v}_t\right)^2 + \frac{\lambda}{2}\left(\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2\right), \qquad \hat{Y}_{it}(0) = \hat{\mathbf{u}}_i^\top \hat{\mathbf{v}}_t \ \text{ for } (i,t) \notin \Omega.$$

The treatment effect estimate is $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$ on the treated cells. Synthetic control is the special case where $\hat{\mathbf{u}}_i$ is constrained to be a convex combination of control units' rows (Problem 6(d) of the homework).

2 Build It (60 min)

Task 1 — the pipeline

Starting from your Lab 3 code:

1. Generate the panel (notebook Part 1). Record the true $\mathbf{Y}(0)$, the true rank, the treatment dates and the true effects — you will grade yourself against them at the end, and only at the end.

2. Fit by SGD for $K \in \{1, \ldots, 6\}$, $\lambda \in \{0, 0.01, 0.1, 1\}$. Select (K, λ) by cross-validation on observed entries: hold out 10% of Ω at random, five times.
3. Refit on all of Ω at the chosen (K, λ); complete the matrix; compute $\hat{\tau}_{it}$ on the treated cells.

Task 2 — uncertainty

Two ways, both required:

1. Placebo. Pretend a never-treated unit was treated at a random date; complete; the "effect" you recover is pure noise. Repeat 50 times for the null distribution of $\hat{\tau}$.
2. Bootstrap. Resample units with replacement (ML I, Lecture 4), refit, recompute the average treatment effect on the treated. Report the standard deviation over 100 resamples.

Task 3 — what CV cannot see

Cross-validation on Ω holds out random untreated cells; the cells you actually need are a block in the lower-right corner. Compare the CV error with the true error on the treated block (now you may look). Explain the gap with the sampling argument of ML I, Lecture 1: the held-out cells are not drawn from the distribution of the cells you predict.

3 Write It Up (40 min)

Deliverable

A notebook and a report of at most 4 pages:

1. The model, the objective, and the assumption that makes completion possible (low rank), stated precisely.
2. The selection procedure and the chosen (K, λ); the CV curve.
3. The estimated effects with placebo and bootstrap uncertainty; one figure.
4. The CV-vs-truth gap and what it implies for a real panel where the truth is unavailable.
5. One derivation from ML II, reproduced in your own words: the SGD gradient for the factorization, the ALS closed form, or the trace-norm connection of Lecture 4.

Rubric (out of 20): model and assumptions 4, selection 4, uncertainty 5, the gap 3, derivation 4.

4 Extension: Self-Attention From Scratch (if time allows)

Problem 18

Implement single-head scaled dot-product self-attention in numpy for a $T \times D$ input: $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ projections, scores $\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k}$, row-softmax, output. Match `torch.nn.MultiheadAttention` with one head to 10^{-6}. Then multi-head with the output projection $\mathbf{W}_O$. Before coding, reproduce Lecture 6's derivation of why the scores are divided by $\sqrt{d_k}$: with i.i.d. unit-variance entries, $\mathrm{Var}(\mathbf{q}^\top \mathbf{k}) = d_k$, and an unscaled softmax over scores of that size saturates.

Submission

Project notebook and PDF, `project_lastname_firstname.*`, before Friday, October 30, 23:59. The oral exam study guide (`reference/deep_learning_up/final_exam/`) covers Lectures 1–6 of ML II; its "Latent Factor Models" and "Attention" sections are the ones this lab prepares you for.